

Displays

D1:

$$\dot{x} = f(x, u) = J(x)\nabla H(x) + G(x)u, \quad (1)$$

D2:

$$\|f(x_1, u) - f(x_2, u)\| \leq L\|x_1 - x_2\|, \forall x_1, x_2 \in \mathcal{X}.$$

D3:

$$\|f(x, u)\| \leq C_f, \forall x \in \mathcal{X}, \forall u \in U. \quad (2)$$

D4:

$$\Sigma_E := \{x \in \mathcal{X} : H(x) = E\}.$$

D5:

$$\phi(t, x_0, u) \in S_{\text{tgt}}.$$

D6:

$$K_\alpha^E := \text{supp}(\mu_\alpha^E) \subseteq \Sigma_E,$$

D7:

$$\overline{\{\phi(t, x, 0) : t \geq 0\}} = K_\alpha^E.$$

D8:

$$\mathcal{A} := \{u_i : (0, \tau_i] \rightarrow U\}_{i=0}^M,$$

D9:

$$\mathcal{K} := \{(x_i, r_i, u_i)\}_{i=1}^N \subseteq \mathbb{R}^n \times \mathcal{R}_{>0} \times \mathcal{A},$$

D10:

$$\text{Supp}(\mathcal{K}) := \bigcup_{i=1}^N \mathcal{B}_{r_i}(x_i),$$

D11:

$$\iota_{\mathcal{K}}(x) := \begin{cases} \arg \min_{1 \leq i \leq N} \frac{\|x - x_i\|}{r_i}, & \rho_{\mathcal{K}}(x) \leq 1, \\ 0, & \text{otherwise.} \end{cases}$$

D12:

$$\pi_{\mathcal{K}} : \mathcal{X} \rightarrow \mathcal{A}$$

D13:

$$u_{\mathcal{K},x}(t) := u_n(t - s_n), \quad t \in [s_n, s_{n+1}).$$

D14:

$$H(S_{\text{tgt}}) = [H_{\min}, H_{\max}],$$

D15:

$$S_{\text{tgt}} \cap K_\alpha^E \neq \emptyset.$$

D16:

$$H_+^\star := \frac{H_{\max} + H_{\min}}{2}, \quad H_-^\star := \frac{H_{\max} - H_{\min}}{2}.$$

D17:

$$\Delta H(x) := |H(x) - H_+^\star| - H_-^\star.$$

D18:

$$H_{\text{tgt}} := \{x \in \mathcal{X} : \Delta H(x) \leq 0\}.$$

D19:

$$H_{\text{tgt}}^\epsilon := \{x \in \mathcal{X} : \Delta H(x) \leq -\epsilon\}.$$

D20:

$$\phi(T, x, u) \in H_{\text{tgt}}^\epsilon.$$

D21:

$$T_\epsilon(x, u) := \inf\{t > 0 : \phi(t, x, u) \in H_{\text{tgt}}^\epsilon\}.$$

$$\text{D22:} \quad v_\epsilon(x, u) := \frac{\Delta H(x) - \Delta H(\phi(T_\epsilon(x, u), x, u))}{T_\epsilon(x, u)}. \quad (3)$$

$$\text{D23:} \quad v_\epsilon(x, u) = \frac{\Delta H(x) + \epsilon}{T_\epsilon(x, u)}.$$

$$\text{D24:} \quad v_\epsilon(x) := \sup_{u \in \mathcal{U}} v_\epsilon(x, u) = \frac{\Delta H(x) + \epsilon}{T_\epsilon^*(x)}, \quad (4)$$

$$\text{D25:} \quad \underline{v}_\epsilon := \inf_{x \in \mathcal{X} \setminus H_{\text{tgt}}^\epsilon} v_\epsilon(x) > 0.$$

$$\text{D26:} \quad \Delta H(x_i(\tau_i)) + v_0 \tau_i + L_H r_i e^{L \tau_i} \leq \Delta H(x_i) - L_H r_i,$$

$$\text{D27:} \quad \Delta H(x) \leq c \implies H(x) \in H(\text{Supp}(\mathcal{K})).$$

$$\text{D28:} \quad \text{int}(\text{Supp}(\mathcal{K})) \cap K_\alpha^E \neq \emptyset.$$

$$\text{D29:} \quad \phi(t, x_0, \pi_{\mathcal{K}}) \in S_{\text{tgt}}.$$

$$\text{D30:} \quad \|\phi(t, x_i, u_i) - \phi(t, y, u_i)\| \leq r_i e^{L t}.$$

$$\text{D31:} \quad \Delta H(\phi(\tau_i, y, u_i)) \leq \Delta H(\phi(\tau_i, x_i, u_i)) + L_H r_i e^{L \tau_i}, \quad (5)$$

$$\text{D32:} \quad \Delta H(x_i) \leq \Delta H(y) + L_H r_i. \quad (6)$$

$$\text{D33:} \quad \Delta H(\phi(\tau_i, y, u_i)) + v_0 \tau_i \leq \Delta H(y). \quad (7)$$

$$\text{D34:} \quad y' := \phi(\tau_i, y, u_i)$$

$$\text{D35:} \quad \Delta H(y') \leq \Delta H(y).$$

$$\text{D36:} \quad \Delta H(y') \leq c := \sup_{x \in S_0} \Delta H(x).$$

$$\text{D37:} \quad \Sigma_E, \quad E := H(y').$$

$$\text{D38:} \quad \phi(T, y', 0) \in \text{Supp}(\mathcal{K}).$$

$$\text{D39:} \quad v_\epsilon(x) \leq L_H C_f, \quad \text{and} \quad T_\epsilon^*(x) \geq \frac{\epsilon}{L_H C_f}.$$

$$\begin{aligned} \text{D40:} \quad & |\Delta H(x) - \Delta H(\phi(T, x, u))| \leq L_H \|x - \phi(T, x, u)\| \\ & = L_H \left\| \int_0^T f(\phi(t, x, u), u(t)) dt \right\| \leq L_H C_f T. \end{aligned}$$

$$\text{D41:} \quad v_\epsilon(x, u) = \frac{\Delta H(x) - \Delta H(\phi(T_\epsilon(x, u), x, u))}{T_\epsilon(x, u)} \leq L_H C_f.$$

D42:

$$v_\epsilon(x) = \frac{\Delta H(x) + \epsilon}{T_\epsilon^*(x)} \leq L_H C_f, \Rightarrow T_\epsilon^*(x) \geq \frac{\Delta H(x) + \epsilon}{L_H C_f}.$$

D43:

$$T_\epsilon(x, u) \geq T_\epsilon^*(x) \geq \frac{\epsilon}{L_H C_f}, \quad \forall x \in \mathcal{X} \setminus H_{\text{tgt}}, \quad u \in \mathcal{U}.$$

D44:

$$\forall E \in H(\text{Supp}(\mathcal{K})), \quad K_\alpha^E = \Sigma_E.$$

D45:

$$H(y) \geq H(x) + \nabla H(x)^\top (y - x) + \frac{\mu_H}{2} \|y - x\|^2.$$

D46:

$$r_i = \frac{(v_\epsilon(x_i) - v_0)T_\epsilon^*(x_i)}{L_H(1 + e^{LT_\epsilon^*(x_i)})} > 0,$$

D47:

$$N \leq (H_2 - H_1) \frac{16L_H^2}{\mu_H(1 - \frac{v_0}{\underline{v}_\epsilon})^2} \frac{\exp(\frac{2L(L_H D_{\mathcal{X}} + \epsilon)}{\underline{v}_\epsilon})}{\epsilon^2}.$$

D48:

$$\mathcal{X}_c := \{x \in \mathcal{X} : \Delta H(x) \leq c\} \setminus H_{\text{tgt}}^\epsilon.$$

D49:

$$\phi(T_\epsilon^*(x), x, u_x) \in \partial H_{\text{tgt}}^\epsilon, \quad v_\epsilon(x) = \frac{\Delta H(x) + \epsilon}{T_\epsilon^*(x)}$$

D50:

$$r(x) := \frac{(v_\epsilon(x) - v_0)T_\epsilon^*(x)}{L_H(1 + e^{LT_\epsilon^*(x)})}.$$

D51:

$$\begin{aligned} & \Delta H(\phi(T_\epsilon^*(x), x, u_x)) + v_0 T_\epsilon^*(x) + L_H r(x) e^{LT_\epsilon^*(x)} \\ &= \Delta H(x) - L_H r(x), \end{aligned}$$

D52:

$$\Delta H(x) + \epsilon - v_0 T_\epsilon(x)^* \geq (\Delta H(x) + \epsilon) \left(1 - \frac{v_0}{\underline{v}_\epsilon}\right) > 0,$$

D53:

$$r(x) \geq \frac{(\Delta H(x) + \epsilon) \left(1 - \frac{v_0}{\underline{v}_\epsilon}\right)}{L_H \left(1 + \exp\left(\frac{L(\Delta H(x) + \epsilon)}{\underline{v}_\epsilon}\right)\right)}$$

D54:

$$\geq \frac{\epsilon \left(1 - \frac{v_0}{\underline{v}_\epsilon}\right)}{2L_H \exp\left(\frac{L(L_H D_{\mathcal{X}} + \epsilon)}{\underline{v}_\epsilon}\right)}, \quad (8)$$

D55:

$$H(x + rv) + H(x - rv) \geq 2H(x) + \mu_H r^2,$$

D56:

$$\max\{H(x + rv), H(x - rv)\} \geq H(x) + \frac{\mu_H}{2} r^2. \quad (9)$$

D57:

$$\begin{aligned} & \max_{y_1, y_2 \in \mathcal{B}_{r(x)}(x)} (H(y_1) - H(y_2)) \\ & \geq \max_{y \in \mathcal{B}_{r(x)}(x)} (H(y) - H(x)) \\ & \geq \frac{\mu_H}{2} r(x)^2. \end{aligned}$$

D58:

$$\begin{aligned} & \max_{y_1, y_2 \in \mathcal{B}_{r(x)}(x)} |\Delta H(y_1) - \Delta H(y_2)| \\ &= \max_{y_1, y_2 \in \mathcal{B}_{r(x)}(x)} |H(y_1) - H(y_2)| \\ &\geq \frac{\mu_H}{2} r(x)^2. \end{aligned}$$

D59:

$$\max\{|H(x) - H_+^*|, |H(x) + \frac{\mu_H}{2} r(x)^2 - H_+^*|\} \geq \frac{\mu_H}{4} r(x)^2.$$

D60:

$$\max_{y_1, y_2 \in \mathcal{B}_{r(x)}(x)} |\Delta H(y_1) - \Delta H(y_2)|$$

D61:

$$\geq \max_{y_1 \in \mathcal{B}_{r(x)}(x)} |H(y_1) - H_+^*| \quad (10)$$

D62:

$$\geq \frac{\mu_H}{4} r(x)^2,$$

D63:

$$\max_{y_1, y_2 \in \mathcal{B}_{r(x)}(x)} |\Delta H(y_1) - \Delta H(y_2)| \geq \frac{\mu_H}{4} r(x)^2$$

D64:

$$\geq \frac{\mu_H (1 - \frac{v_0}{v_\epsilon})^2}{16L_H^2} \frac{\epsilon^2}{\exp(\frac{2L(L_H D_{\mathcal{X}} + \epsilon)}{v_\epsilon})} \quad (11)$$

D65:

$$[H_1, H_2] := H(\mathcal{X}_c) \cup H(H_{\text{tgt}}^\epsilon).$$

D66:

$$\begin{aligned} & \max_{y_1, y_2 \in \mathcal{B}_{r(x)}(x)} |H(y_1) - H(y_2)| \geq \frac{\mu_H}{4} r(x)^2 \\ & \geq \frac{\mu_H (1 - \frac{v_0}{v_\epsilon})^2}{16L_H^2} \frac{\epsilon^2}{\exp(\frac{2L(L_H D_{\mathcal{X}} + \epsilon)}{v_\epsilon})} \end{aligned}$$

D67:

$$[H_1, H_2] \subseteq \bigcup_{i=1}^N \mathcal{B}_{r_i}(x_i).$$

D68:

$$\mathcal{K} := \{(x_i, r_i, u_i)\}_{i=1}^N,$$

D69:

$$[H_1, H_2] \subseteq H(\text{Supp}(\mathcal{K})),$$

D70:

$$\phi(t, x_0, \pi_{\mathcal{K}}) \in S_{\text{tgt}}.$$

D71:

$$\begin{aligned} N &\leq \frac{H_2 - H_1}{\frac{\mu_H (1 - \frac{v_0}{v_\epsilon})^2}{16L_H^2} \frac{\epsilon^2}{\exp(\frac{2L(L_H D_{\mathcal{X}} + \epsilon)}{v_\epsilon})}} \\ &= (H_2 - H_1) \frac{16L_H^2}{\mu_H \left(1 - \frac{v_0}{v_\epsilon}\right)^2} \frac{\exp\left(\frac{2L(L_H D_{\mathcal{X}} + \epsilon)}{v_\epsilon}\right)}{\epsilon^2}. \end{aligned}$$

D72:

$$T_{\max} \leq \frac{L_H D_{\mathcal{X}}}{v_0} \left(1 + \frac{T_1}{\tau_{\min}}\right) + T_2.$$

D73:

$$y_i = \phi(t_{1,i}, x_i, 0) \in \text{Supp}(\mathcal{K}).$$

D74:

$$x_{i+1} = \phi(t_{2,i}, y_i, u_i),$$

D75:

$$\Delta H(x_{i+1}) + v_0 t_{2,i} \leq \Delta H(y_i),$$

D76:

$$N\tau_{\min} \leq \sum_{i=1}^N t_{2,i} \leq \frac{\Delta H(x_0)}{v_0},$$

D77:

$$N \leq \left\lfloor \frac{\Delta H(x_0)}{v_0 \tau_{\min}} \right\rfloor.$$

D78:

$$\forall x_0 \in S_0, \exists T \leq \bar{T} \text{ s.t. } \phi(T, x_0, \pi_{\mathcal{K}}) \in H_{\text{tgt}},$$

D79:

$$\bar{T} \leq \sum_{i=1}^N t_{2,i} + NT_1 \tag{12}$$

D80:

$$\leq \frac{\Delta H(x_0)}{v_0} + \left\lfloor \frac{\Delta H(x_0)}{v_0 \tau_{\min}} \right\rfloor T_1 \tag{13}$$

D81:

$$\leq \frac{L_H D_{\mathcal{X}}}{v_0} \left(1 + \frac{T_1}{\tau_{\min}}\right). \tag{14}$$

D82:

$$T_{\max} \leq \bar{T} + T_2 \leq \frac{L_H D_{\mathcal{X}}}{v_0} \left(1 + \frac{T_1}{\tau_{\min}}\right) + T_2.$$

D83:

$$r_i(t) = \frac{\Delta H(x_i) - \Delta H(\phi(t, x_i, u_{i,t})) - v_0 t}{L_H + L_H e^{Lt}}.$$

D84:

$$\begin{aligned} x &= \begin{bmatrix} q \\ p \end{bmatrix} = \begin{bmatrix} q \\ m\dot{q} \end{bmatrix}, H(x) = \frac{p^2}{2m} + \frac{k}{2}q^2, \\ \dot{x} &= \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} \nabla H(x) + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u. \end{aligned}$$

D85:

$$\begin{aligned} x &= \begin{bmatrix} q \\ p \end{bmatrix} = \begin{bmatrix} \theta \\ m\ell^2\dot{\theta} \end{bmatrix}, H(x) = \frac{p^2}{2m\ell^2} + mg\ell(1 - \cos q), \\ \dot{x} &= \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} \nabla H(x) + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u. \end{aligned}$$

Inline

I1: $\|\cdot\|$

I2: $x \in \mathbb{R}^n$

I3: $r > 0$

I4: $\mathcal{B}_r(x) := \{w \in \mathbb{R}^n \mid \|x - w\| \leq r\}$

I5: r

I6: x

I7: $S \subseteq \mathbb{R}^n$

I8: $D_S := \sup_{x,y \in S} \|x - y\|$

I9: $\overline{S}$

I10: ∂S
 I11: $\text{int}(S)$
 I12: $\forall x \in \mathbb{R}^n$
 I13: S
 I14: $d(x, S) := \inf_{y \in S} \|x - y\|$
 I15: $\lceil \cdot \rceil$
 I16: $\lfloor \cdot \rfloor$
 I17: $x \in \mathcal{X} \subset \mathbb{R}^n$
 I18: $\mathcal{X}$
 I19: $H : \mathbb{R}^n \rightarrow \mathbb{R}$
 I20: $J(x) \in \mathbb{R}^{n \times n}$
 I21: $u \in U \subset \mathbb{R}^m$
 I22: $H(x)$
 I23: $J(x)$
 I24: H
 I25: $L_H > 0$
 I26: $\forall x \in \mathcal{X}, \|\nabla H(x)\| \leq L_H$
 I27: $f(x, u)$
 I28: $u \in U$
 I29: $L > 0$
 I30: U
 I31: f
 I32: $C_f > 0$
 I33: $H(\cdot)$
 I34: $E \in H(\mathcal{X})$
 I35: Σ_E
 I36: $\mathcal{U}^{(0, t]}$
 I37: $(0, t]$
 I38: $u : (0, t] \rightarrow U$
 I39: $\mathcal{U} := \mathcal{U}^{(0, \infty)}$
 I40: $x \in \mathcal{X}$
 I41: $u \in \mathcal{U}^{(0, t]}$
 I42: $\phi(t, x, u)$
 I43: t
 I44: $S_0 \subseteq \mathcal{X}$
 I45: $S_{\text{tgt}} \subseteq \mathcal{X}$
 I46: $x_0 \in S_0$
 I47: S_{tgt}
 I48: $u \in \mathcal{U}^{(0, t]}$
 I49: $t > 0$
 I50: $E \in H(\mathcal{X})$
 I51: $\Sigma_E := \{x \in \mathcal{X} : H(x) = E\}$
 I52: $\phi(t, x, 0)$
 I53: μ
 I54: M
 I55: ϕ
 I56: $A \subseteq M$
 I57: $t \geq 0$
 I58: $\mu(\phi(t, A)) = \mu(A)$.
 I59: $\phi(t, A, 0) \subseteq A$
 I60: $\mu(A) \in \{0, 1\}$
 I61: μ_E
 I62: $\{\mu_\alpha^E\}_{\alpha \in \mathcal{A}_E}$
 I63: ν_E
 I64: $\mathcal{A}_E$
 I65: $\mu_E = \int_{\mathcal{A}_E} \mu_\alpha^E d\nu_E(\alpha)$

I66: $\alpha \in \mathcal{A}_E$
 I67: $\text{supp}(\mu) := \{x \in \Sigma_E : \mu(\mathcal{B}_r(x)) > 0, \forall r > 0\}$
 I68: μ_α^E
 I69: K_α^E
 I70: $x \in K_\alpha^E$
 I71: $\mathcal{D} := \{(x_j, u_j(\cdot), \tau_j)\}_{j=1}^M$,
 I72: $u_j : (0, \tau_j] \rightarrow U$
 I73: x_j
 I74: $u_0 : (0, \tau_0] \rightarrow U$
 I75: $\tau_0 > 0$
 I76: u_i
 I77: $\tau_i > 0$
 I78: $x_i \in \mathbb{R}^n$
 I79: $u_i \in \mathcal{A}$
 I80: $r_i > 0$
 I81: $\mathcal{K}$
 I82: $N := |\mathcal{K}|$
 I83: $\text{Supp}(\mathcal{K})$
 I84: $\rho_{\mathcal{K}}(x) := \min_{1 \leq i \leq N} \frac{\|x - x_i\|}{r_i}$.
 I85: $\iota_{\mathcal{K}} : \mathcal{X} \rightarrow \{0, 1, \dots, N\}$
 I86: $x \in \text{Supp}(\mathcal{K})$
 I87: u_0
 I88: $\pi_{\mathcal{K}}(x) = u_{\iota_{\mathcal{K}}(x)}$.
 I89: $x_0 = x$
 I90: $\pi_{\mathcal{K}}$
 I91: $n \geq 0$
 I92: $u_n := \pi_{\mathcal{K}}(x_n), T_n := \tau(u_n), x_{n+1} := \phi(T_n, x_n, u_n)$
 I93: $\tau(u_n)$
 I94: u_n
 I95: $s_0 := 0$
 I96: $s_{n+1} := s_n + T_n$
 I97: $u_{\mathcal{K},x} : (0, \infty) \rightarrow U$
 I98: $H_{\min} := \min_{x \in S_{\text{tgt}}} H(x)$
 I99: $H_{\max} := \max_{x \in S_{\text{tgt}}} H(x)$.
 I100: $E \in [H_{\min}, H_{\max}]$
 I101: $K_\alpha^E \subseteq \Sigma_E$
 I102: $H(S_{\text{tgt}}) = [H_{\min}, H_{\max}]$
 I103: $\Delta H(x) \leq 0$
 I104: $H(x) \in H(S_{\text{tgt}})$
 I105: $0 < \epsilon < H_-^*$
 I106: H_{tgt}^ϵ
 I107: $x \in \mathcal{X} \setminus H_{\text{tgt}}^\epsilon$
 I108: $T > 0$
 I109: $u \in \mathcal{U}^{(0,T]}$
 I110: $x \in \mathcal{X}$
 I111: $u \in \mathcal{U}$
 I112: $\Delta H(x)$
 I113: $\phi(T_\epsilon(x, u), x, u) \in \partial H_{\text{tgt}}^\epsilon$
 I114: $T_\epsilon^*(x)$
 I115: $T_\epsilon^*(x) := \inf_{u \in \mathcal{U}} T_\epsilon(x, u)$
 I116: $\epsilon > 0$
 I117: $\mathcal{K} = \{(x_i, r_i, u_i)\}_{i=1}^N$
 I118: $(x_i, r_i, u_i) \in \mathcal{K}$
 I119: $x_i(\tau_i) := \phi(\tau_i, x_i, u_i)$
 I120: $\tau_{\min} = \min_i \tau_i$, and $v_0 > 0$
 I121: $c := \sup_{x \in S_0} \Delta H(x)$

I122: $E \in H(\text{Supp}(\mathcal{K}))$
 I123: $t < \infty$
 I124: $y \in \text{Supp}(\mathcal{K})$
 I125: i
 I126: $y \in \mathcal{B}_{r_i}(x_i)$
 I127: $\|y - x_i\| \leq r_i$
 I128: $\|\nabla H(x)\| \leq L_H$
 I129: ΔH
 I130: L_H
 I131: $y \in \text{Supp}(\mathcal{K})$
 I132: S_0
 I133: $H(y') \in H(\text{Supp}(\mathcal{K}))$
 I134: y'
 I135: $u \equiv 0$
 I136: $\text{int}(\text{Supp}(\mathcal{K}))$
 I137: $K_\alpha^E \subseteq \Sigma_E$
 I138: $y' \in \Sigma_E$
 I139: $\phi(t, y', 0)$
 I140: $T > 0$
 I141: $y \in H_{\text{tgt}}$
 I142: $H(y) \in [H_{\min}, H_{\max}]$
 I143: $\Sigma_{H(y)}$
 I144: $T' > 0$
 I145: $\phi(T', y, 0) \in S_{\text{tgt}}$
 I146: $v_0 \tau_{\min} > 0$
 I147: $N_* = \left\lceil \frac{c}{v_0 \tau_{\min}} \right\rceil$
 I148: H_{tgt}
 I149: $\phi(t, x_0, \pi_{\mathcal{K}}) \in S_{\text{tgt}}$
 I150: $\square$
 I151: n
 I152: α
 I153: $x \in \mathcal{X} \setminus H_{\text{tgt}}$
 I154: $v_\epsilon(x) \leq L_H C_f$
 I155: $\Delta H(x) > 0$
 I156: $x \in \mathcal{X} \setminus H_{\text{tgt}}$
 I157: $\mu_H > 0$
 I158: $x, y \in \mathcal{X}$
 I159: $v_0 \in (0, v_\epsilon)$
 I160: $\mathcal{K} = \{(x_i, r_i, u_i)\}_{i=1}^N$
 I161: $u_i : (0, T_\epsilon^*(x_i)] \rightarrow U$
 I162: $\partial H_{\text{tgt}}^\epsilon$
 I163: x_i
 I164: $T_\epsilon^*(x_i)$
 I165: $x_0 \in S_0$
 I166: $\phi(t, x_0, \pi_{\mathcal{K}}) \in S_{\text{tgt}}$
 I167: $H(\text{Supp}(\mathcal{K})) = [H_1, H_2]$
 I168: $x \in \mathcal{X}_c$
 I169: $u_x : (0, T_\epsilon^*(x)] \rightarrow U$
 I170: $u_x : (0, T_\epsilon^*(x)] \rightarrow U$
 I171: $v_0 < v_\epsilon \leq v_\epsilon(x)$
 I172: $r(x) > 0$
 I173: $(x, r(x), u_x)$
 I174: $v_\epsilon(x) \geq \underline{v}_\epsilon > 0, v_\epsilon(x) = \frac{\Delta H(x_i) + \epsilon}{T_\epsilon^*(x)},$
 I175: $T_\epsilon^*(x) \leq \frac{\Delta H(x) + \epsilon}{\underline{v}_\epsilon}$
 I176: $r = \frac{\Delta H(x) + \epsilon - v_0 T_\epsilon^*(x)}{L_H (1 + e^{L T_\epsilon^*(x)})}$

I177: $0 < v_0 < \underline{v}_\epsilon$
 I178: $r(x)$
 I179: $x \in \mathcal{X}_c$
 I180: $\mathcal{B}_{r(x)}(x)$
 I181: $\|v\| = 1$
 I182: $x \pm rv$
 I183: y_1, y_2
 I184: $y \in \mathcal{B}_{r(x)}(x)$
 I185: $H_+^* \notin H(\mathcal{B}_{r(x)}(x))$
 I186: $H_+^* \in H(\mathcal{B}_{r(x)}(x))$
 I187: $y^* \in \mathcal{B}_{r(x)}(x)$
 I188: $H(x) + \frac{\mu_H}{2}r(x)^2 \leq H(y^*) \leq \max H(\mathcal{B}_{r(x)}(x))$
 I189: $[H(x), H(x) + \frac{\mu_H}{2}r(x)^2] \subseteq H(\mathcal{B}_{r(x)}(x))$
 I190: $y_1, y_2 \in \mathcal{B}_{r(x)}(x)$
 I191: $y_2 \in \mathcal{B}_{r(x)}(x)$
 I192: $H(y_2) = H_+^*$
 I193: $\forall x \in \mathcal{X}_c$
 I194: $E \in [H_1, H_2]$
 I195: $H(x) = E$
 I196: $E \in H(\mathcal{B}_{r(x)}(x))$
 I197: $\{H(\mathcal{B}_{r(x)}(x))\}_{x \in \mathcal{X}_c}$
 I198: $[H_1, H_2]$
 I199: $r_i := r(x_i), u_i := u_{x_i},$
 I200: $\mathcal{B}_{r_i}(x_i)$
 I201: η
 I202: $\forall x$
 I203: $H(x) \in [H_1, H_2]$
 I204: $T_1 > 0, \text{ s.t.}$
 I205: $\min_{t \in (0, T_1]} d(\phi(t, x, 0), \text{Supp}(\mathcal{K})) = 0$
 I206: $H(x) \in [H_{\min}, H_{\max}]$
 I207: $T_2 > 0, \text{ s.t.}$
 I208: $\min_{t \in (0, T_2]} d(\phi(t, x, 0), S_{\text{tgt}}) = 0$
 I209: $\tau_{\min} := \min_i \tau_i$
 I210: $x \in S_0$
 I211: T_1
 I212: $\{x_i\}_{i=0, \dots, N}$
 I213: $\{y_i\}_{i=0, \dots, N}$
 I214: $0 \leq t_{1,i} \leq T_1$
 I215: y_i
 I216: $u_i \in \mathcal{U}^{(0, \tau_i]}$
 I217: $t_{2,i} := \tau_i$
 I218: $i = 0, \dots, N$
 I219: $x_i \xrightarrow{t_{1,i}} y_i \xrightarrow{t_{2,i}} x_{i+1}$
 I220: N
 I221: $x_N \in H_{\text{tgt}}^\epsilon$
 I222: $t_{1,i} \leq T_1$
 I223: $\overline{T}$
 I224: $T_{\max}$
 I225: $\mathcal{D} = \{(x_j, u_j(\cdot), T_j)\}_{j=1}^M$
 I226: $(x_j, u_j(\cdot), T_j) \in \mathcal{D}$
 I227: $\phi(t, x_j, u_j)$
 I228: $t \in (0, T_j]$
 I229: $x_j \in S_0$
 I230: $\phi(T_j, x_j, u_j) \in S_{\text{tgt}}^\delta, \forall j = 1, \dots, M$
 I231: $S_{\text{tgt}}^\delta \subseteq H_{\text{tgt}}^\epsilon \cap S_{\text{tgt}}$
 I232: $s \in [0, T_j)$

I233: $x_i := \phi(s, x_j, u_j)$.
 I234: v_0
 I235: $t \in (0, T_j - s]$
 I236: $u_{i,t}$
 I237: u_j
 I238: $(s, s + t]$
 I239: $\phi(t, x_i, u_{i,t}) = \phi(s + t, x_j, u_j)$
 I240: $r_i(t) > 0$
 I241: $\mathcal{B}_{r_i(t)}(x_i)$
 I242: $t_i \in \arg \max_{t \in (0, T_j - s], r_i(t) > 0} r_i(t)$
 I243: $\tau_i := t_i$
 I244: $u_i := u_{i,t_i}$
 I245: $r_i := r_i(t_i)$
 I246: (x_i, r_i, u_i)
 I247: $\sigma_i := \inf\{\delta \in (0, \tau_i] \mid \phi(s + \delta, x_j, u_j) \in \partial \mathcal{B}_{r_i}(x_i)\}$,
 I248: $\sigma_i := \tau_i$
 I249: $x_{i+1} := \phi(s + \sigma_i, x_j, u_j)$,
 I250: $\mathcal{D}$
 I251: $\mathcal{K} = \{(x_i, r_i, u_i)\}_{i=1}^N$.
 I252: $\forall x \in \mathcal{X}$
 I253: $\mathcal{K} \leftarrow \emptyset$
 I254: $j = 1, \dots, M$
 I255: $s \leftarrow 0$
 I256: $s < T_j$
 I257: $\phi(s, x_j, u_j) \notin S_{\text{tgt}}$
 I258: $x_i \leftarrow \phi(s, x_j, u_j)$
 I259: t_i
 I260: $\tau_i \leftarrow t_i$
 I261: $u_i \leftarrow u_{i,t_i}$
 I262: $r_i \leftarrow r_i(t_i)$
 I263: $\mathcal{K} \leftarrow \mathcal{K} \cup \{(x_i, r_i, u_i)\}$
 I264: $\sigma_i \leftarrow \inf\{\delta : \|\phi(s + \delta, x_j, u_j) - x_i\| = r_i\}$
 I265: $\sigma_i \leftarrow \tau_i$
 I266: $s \leftarrow s + \sigma_i$
 I267: $x = [q^\top, p^\top]^\top$
 I268: q
 I269: p
 I270: x^*
 I271: $S_{\text{tgt}} = \{x : \|x - x^*\| \leq \varepsilon\}$
 I272: $\varepsilon = 0.1$
 I273: $(24, 24, 16)$
 I274: 1.2×10^{-3}
 I275: 5×10^{-4}
 I276: $\bar{H}$
 I277: $m = 1$
 I278: $k = 1$
 I279: $u \in [-20, 20]$
 I280: $x^* = [0, 0]^\top$
 I281: $M = 1$
 I282: $M = 2$
 I283: $M = 3$
 I284: $\ell = 2$
 I285: $g = 9.81\text{m/s}^2$
 I286: $x^* = [\pi, 0]^\top$
 I287: $M \geq 3$
 I288: $M = 1, \dots, 5$

I289: $M = 5$